higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

INSTRUMENTATION TECHNOLOGY
NQF LEVEL 3

12 NOVEMBER 2019

This marking guideline consists of 9 pages.

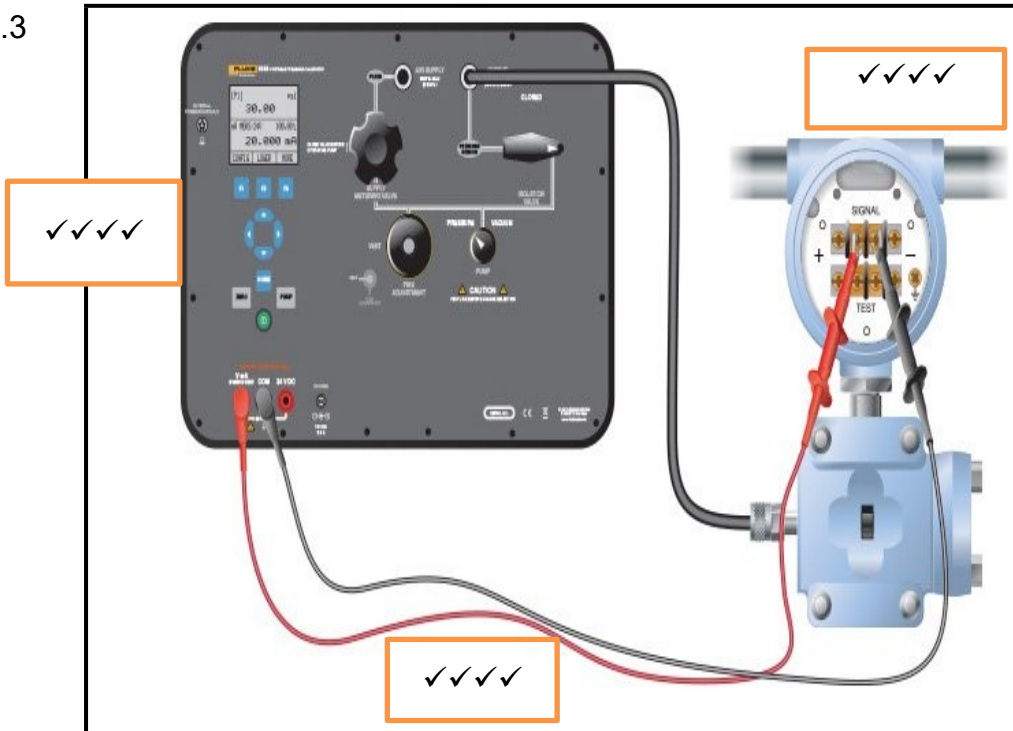
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QUESTION 1

1.1 1.1.1 The bench provides a stable ambient environment✓ for calibration, an opportunity to use the most accurate equipment and protection from factory conditions✓ during the commission testing and calibration of pressure transmitters.✓ (3)

1.1.2
- Connect the transmitter test hose from the calibrator to the transmitter.
- Connect the mA measurement jacks of the calibrator to the transmitter.
- Set the pressure/vacuum selection knob to the necessary function.
- Close the vent knob and supply metering valve.
- Apply pressure or vacuum from the pump by holding down the pump bottom and release when the necessary pressure is reached.
- Correct the pressure with the fine pressure adjustment.
- Read the reference pressure and the current output of the transmitter from the display.
- Repeat all test points. If the measured mA signal at the test points is found within tolerance the test is complete. If not, then adjustment is required. (8)

1.1.3



4 marks for the showing of measuring cables
4 marks for the calibrating instrument
4 marks for the measuring instrument (12)

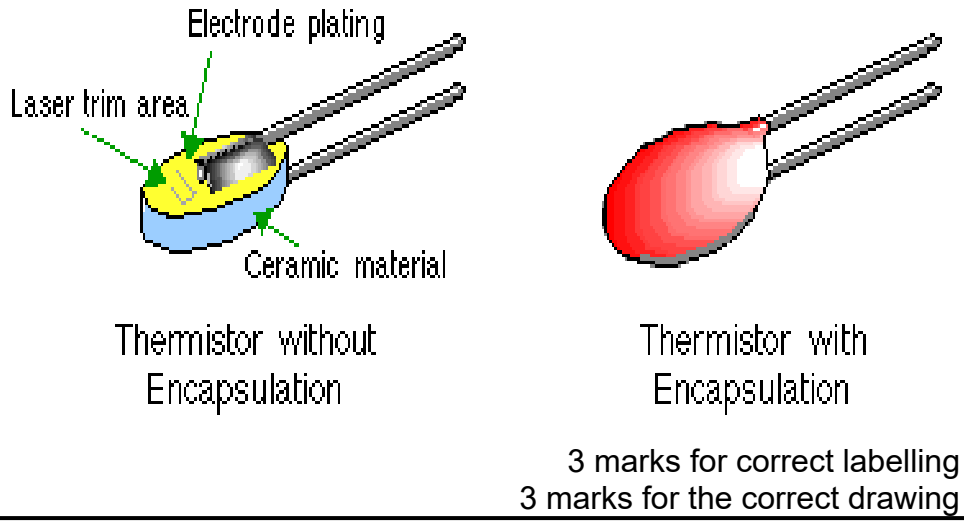
[23]

QUESTION 2

2.1 2.1.1

- It is the short form for thermal resistor, which is a thermally sensitive resistor.
- A temperature-sensing element. (Any 1 × 2) (2)

2.1.2



3 marks for correct labelling
3 marks for the correct drawing (6)

2.1.3

- A thermistor is used in control devices actuated by temperature.
- It is used in automatic temperatures controllers.
- It is widely used in temperature measurement.
- In electronics circuits used for temperature compensation.
- It is also used for biasing and compensating circuits of transistors.
- Terminators used as temperature sensors with or without compensation.
- Used for the measurement of thermal conductivity.
- Used for the measurement of RF power.
- Used in thermal relay circuits and time delay circuits.
- Thermistors are used in measuring temperature distribution or temperature gradient. (Any 5 × 1) (5)

2.1.4 A thermistor consists of a ceramic-like semi-conducting material✓ such as copper oxide, manganese oxide, nickel oxide, cobalt oxide, titanium oxide and lithium oxide.✓ These oxides are blended in suitable proportions and compressed into desired shapes✓ and heat treated to recrystallise them,✓ resulting in a dense ceramic body with the required temperature resistance characteristics.✓ (5)

2.1.5 As the temperature increases, resistance decreases and vice versa✓ and this is called negative temperature characteristics of thermocouples. Thermistors are nonlinear instruments✓ whereas resistors such as platinum, copper and tungsten are quite linear.✓ (3)

2.1.6
(a) In thermistors, semiconductors or ceramic materials such as manganese, cobalt, uranium, etc. are used.
(b) Have low hysteresis
(c) Low cost
(d) Are larger in size
(e) Large and nonlinear changes in resistance
(f) Semiconductors are used which result in changes in resistance with change in temperature
(g) The reproducibility is good
(h) The range of measurements are -50 to 3 000 °C

(8)

2.2 2.2.1 Pyrometer (1)

2.2.2 A pyrometer may be calibrated by sighting it through a hole into a block body enclosure.✓ A special furnace was built for this purpose. Two thermocouples are placed in the furnace, one in the front and one at the back.✓ The two temperatures are compared until the oven is at a uniform temperature and then the pyrometer is sighted into the furnace and the reading is taken.✓ The thermocouple reading is then compared with the pyrometer reading.✓ (4)

2.2.3 If two bodies of different temperatures are placed opposite each other,✓ transfer of heat energy from the hot body to the colder body takes place,✓ and the amount of radiated energy per time unit is roughly proportional to the fourth power of the absolute temperature difference between the two bodies.✓ (3)

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QUESTION 3

3.1 3.1.1

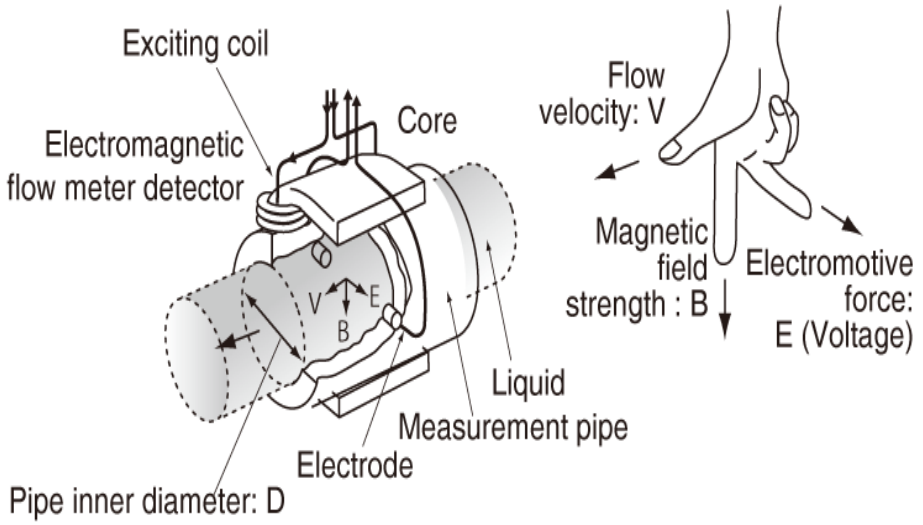
- The substance being measured must be conductive. Therefore, it can't be employed for metering the flow rate of gases and steam, petroleum products and similar liquids with very low conductivity.
- To render the meter insensitive to variations in the resistance of a liquid, the effective resistance of the liquid between the electrodes should not exceed 1% of the impedance of the external circuit.
- It is a very expensive device.
- As the meter always measures the volume rate, the volume of any suspended matter in the liquid will be included.
- To avoid any trouble caused by entrained air, when the flow tube is installed in a horizontal pipe-line, the electrodes should be on the horizontal diameter.
- As a zero check on the installation can be performed only by stopping the flow, isolating valves are required, and a bypass may also be necessary through which the flow may be directed during a zero check. The pipe must run full in case regulating valves are installed upstream of the meter. (Any 5 × 1) (5)

3.1.2

- The obstruction to the flow is almost nil and therefore this type of meter can be used for measuring heavy suspensions, including mud, sewage and wood pulp.
- There is no pressure head loss in this type of flow meter other than that of the length of the straight pipe which the meter occupies.
- They are not adversely affected by upstream flow disturbances.
- They are practically unaffected by variations in density, viscosity, pressure, and temperature.
- Electric power requirements can be low (15 or 20 W), particularly with pulsed DC types.
- These meters can be used as bidirectional meters.
- The meters are suitable for most acids, bases, water, and aqueous solutions because the lining materials selected are not only good electrical insulators, but also are corrosion resistant.
- The meters are widely used for slurry services not only because they cannot be obstructed, but also because some of the liners such as polyurethane, neoprene and rubber have good abrasion or erosion resistance.
- They are capable of handling extremely low flows. (5)

3.1.3 Electromagnetic or magnetic induction is the production of an electromotive force✓ across an electrical conductor✓ in a changing magnetic field✓ (3)

3.1.4



5 marks for the correct drawing
3 marks for any three correct labels (8)

3.1.5
- Water supply network
- Waste-water management
- Fertiliser industries
- Paper and pulp industries
- Beverage industries (Any 3 × 1) (3)

3.2
- Low flow cut off
- Relative low power consumption
- Corrosion-resistant
- Bi-directional flow capability
- Pressure drop equals to an equivalent length of straight pipe (5)

3.3 Vortex meters can be installed vertically, horizontally, or at any angle,✓ as long as the pipe is kept full. Installing the meter in a vertical line with upward flow will always keep the pipe full.✓ When the flow is horizontal or downward in a vertical line, keep the downstream piping elevated to trap the fluid.✓
Use check valves to keep the piping full of liquid in the no-flow condition. If the replacement of the meter in a particular piping configuration requires stopping the flow, block and bypass valves can be installed around the meter.✓
Mating flanges (on Schedule 40 or Schedule 80 mating piping) must have the same diameter and smooth bore as the vortex flowmeter.✓ Use weld-neck flanges rather than reducing flanges. Ensure that the inner surface of the mating pipe is free from mill scale, pits, holes, reaming scores, and bumps for a distance of FOUR pipe diameters upstream and TWO pipe diameters downstream of the meter. The bores of the meter, gaskets, and adjacent piping must be carefully aligned to eliminate any obstructions or steps. (6)

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3.4	3.4.1	True		
	3.4.2	True		
	3.4.3	True		
	3.4.4	False		
	3.4.5	False		
			(5 × 1)	(5)
3.5	3.5.1	easy		
	3.5.2	no		
	3.5.3	less		
	3.5.4	wide		
	3.5.5	low		
			(5 × 1)	(5)
				[45]

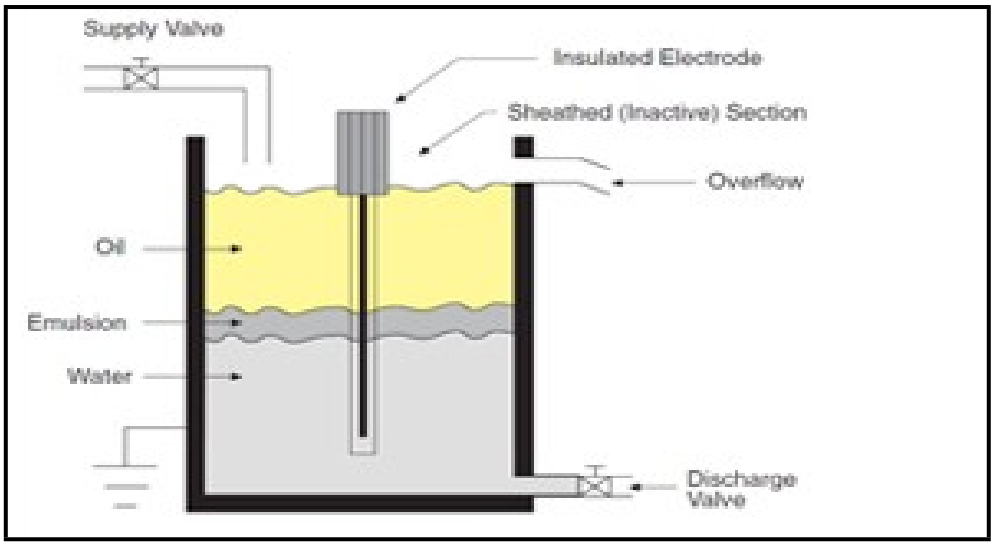
QUESTION 4

4.1 At 4 mA
H = 0
L = 100" × 1.1 = 110" H_2O✓
So that
H – L + B = 0✓
0 – 110 + B = 0✓
B = 110✓

At 20 mA
H = 100. × 1.0 = 100" H_2O✓
L = 100. × 1.1 = 110H_2O

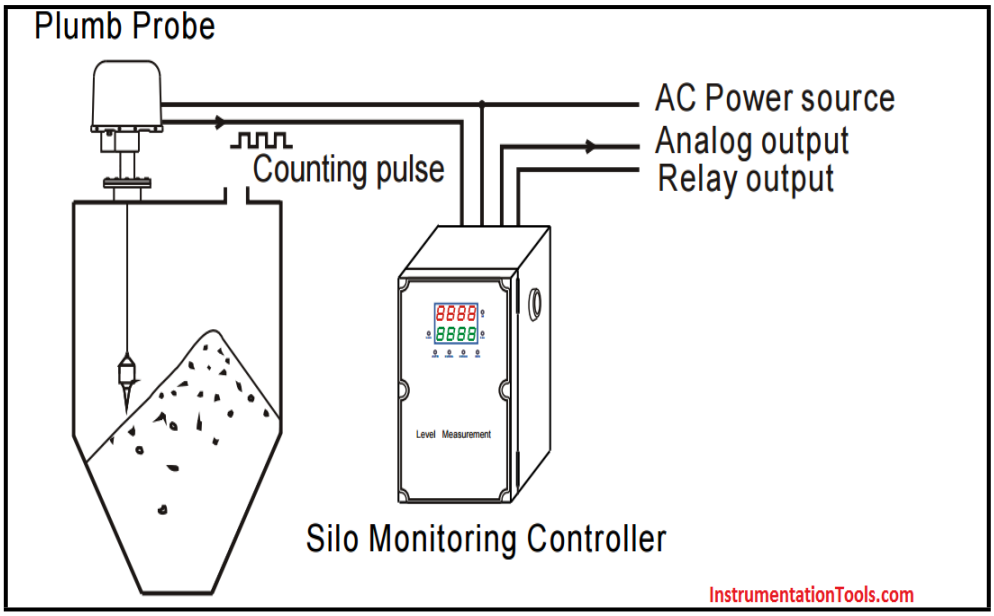
So that
H – L + B = S✓
100 – 110 + 110 = S✓
S 100" H2O✓
So calibration is (-B) to (-B + S)
-110 to (-110 + 100) ✓
Calibration range is 100 to 10" H_2O✓ (10)

4.2 4.2.1



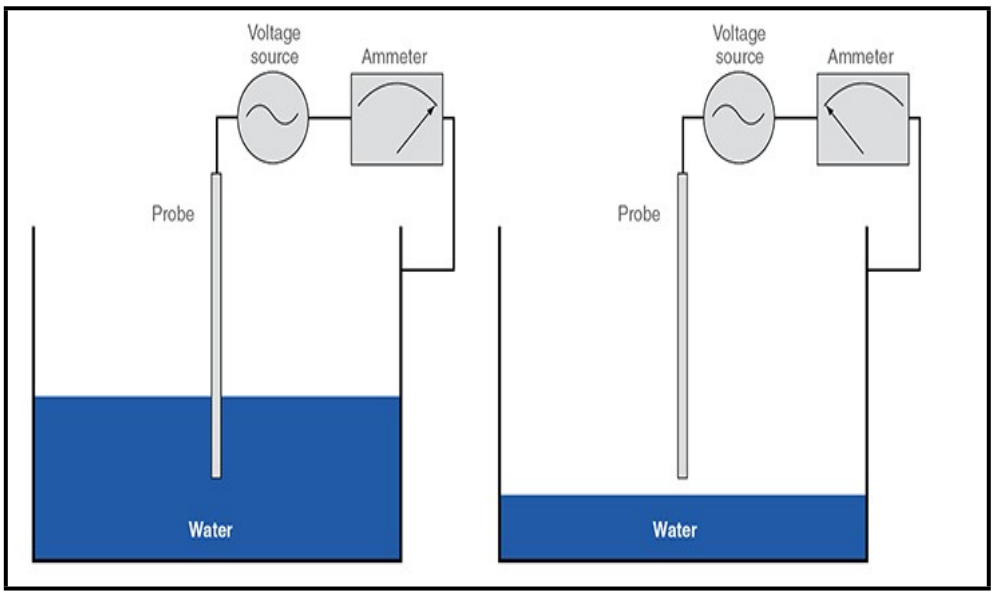
(7)

4.2.2



(7)

4.2.3



(7)

4.3 A range where the value of the measured variable zero✓ is less than✓ the lower range value.✓ The zero does not appear on the scale.✓ (4)

4.4 4.4.1 The importance of level measurement cannot be overstated. Incorrect or inappropriate measurements can cause levels in vessels to be excessively higher or lower than their measured values.✓ Low levels can cause problems and damage equipment,✓ while high levels can cause overflow and potentially create safety and environmental problems.✓ When working in industrial settings, getting accurate level measurements saves time and money,✓ and optimizes the performance of the plant and its processes.✓ (5)

4.4.2
- Capacitance
- Hydrostatic
- Magnetic
- Radar
- Ultrasonic (5)

[45]

TOTAL: 150

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